

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 14
Matrix-Vector Multiplication.

Review

1. What is the *definition* of $A\vec{x}$, for an $m \times n$ matrix A and a $n \times 1$ vector $\vec{x}$?
2. Use the *row interpretation* of matrix-vector multiplication to find $A\vec{x}$.
3. Use the *column interpretation* of matrix-vector multiplication to find $A\vec{x}$.
4. Find $A\vec{e}_1$.
5. Find $A\vec{e}_2$.
6. What is I in $I\vec{x}$?
7. Find $I\vec{x}$.

New (Chp. 6)

1. Find

$$\begin{bmatrix} \cos(45^\circ) & -\sin(45^\circ) \\ \sin(45^\circ) & \cos(45^\circ) \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Linear Functions (Chp. 8)

A be the 3×2 matrix: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

We'll see that $f(\vec{x}) = A\vec{x}$ defines a *linear* function $f: \mathbb{R}^2 \rightarrow \mathbb{R}^3$.

We first check that f is additive.

2. Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Find $A\vec{x}$.

3. Let $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$. Find $A\vec{y}$.

